METRIC
DOD-C-70514 (AR)
23 August 1985

MILITARY SPECIFICATION

CARTRIDGE, 120MM, SUPER SLUG, LOAD, ASSEMBLE, AND PACK

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification establishes the requirements and quality assurance provisions for loading, assembling and packing one type of cartridge designated as Cartridge, 120mm, Super Slug.

1.2 Intended use. - The cartridge is intended to be used for proofing recoil systems and gun/mount combinations for the M1A1 Tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. - Unless otherwise specified, the following specifications, standards, and handbooks of the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Packaging, Methods of
MIL-B-117	Bags, Sleeves and Tubing-Interior Packing
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286	Propellants, Sampling, Inspection and Testing
MIL-STD-1168	Ammunition Lot Numbering
MIL-STD-1169	Packaging, Packing and Marking of Inert Ammunition Data Card
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. - The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

12524837	Rubber, Silicon
12524967	Gage, Profile and Alignment
12526309	Cartridge, 120mm, Super Slug
12527256	Packing and Marking for Propulsion/Ignition Assembly, 120mm Super Slug

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. - The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E8	Tension Testing of Metallic Material
ASTM E10	Brinell Hardness of Metallic Materials, Test For

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ASTM E18	Rockwell Hardness of Metallic Materials, Test For
ASTM D792	Specific Gravity and Density of Plastics by Displacement, Tests For
ASTM D2240	Rubber Property - Durometer Hardness, Test For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ARMY REGULATIONS AR-700-15	Packaging of Material
ATI 139981	Effective Velocity of Escape of the Powder Gas from a Gun

(Application for copies should be addressed to the Defense Technical Center Defense Logistics Agency, Cameron Station, Alexandria, Virginia 22304-6145)

2.3 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing 12526309 and the drawings and specifications referenced therein.

3.3 Propellant weight. - The weight of the M30 propellant within an individual cartridge shall not differ by more than plus or minus 6 grams from that designated for the particular lot of propellant used.

3.4 Performance. - The cartridge shall comply with the following requirements.

3.4.1 Chamber pressure. - The chamber pressure, when fired at 21°C, shall not exceed 6700 bars.

3.4.2 Muzzle velocity. - The average muzzle velocity shall be recorded for information only.

3.4.3 Recoil momentum. - When fired at 21°C, the cartridge shall generate a recoil momentum of 34.3 - 3.0 Kilo Newton seconds.

3.4.4 Misfire. - The cartridge shall not fail to fire from the weapon.

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3.4.5 Metal parts security. - There shall be no indication of parts failure in the gun bore or in flight. Discarding of the Seal-12526248, is not a cause for rejection.

3.4.6 Case base extraction. - After firing, the case base and seal assembly shall be extracted by the service extractors when the weapon is operated in the normal manner. (See 6.5.)

3.5 Adhesive bond. - The test sample cartridge case/cap case shall be acceptable if it separates at other than the glue joint interface under a tension of 8900 Newtons (N) minimum.

3.6 Primer electrical resistance. - The primer electrical resistance shall not exceed the following:

Primer electrode to primer body	2.5 ohms maximum, 1.0 ohms min.
Primer body to case base	0.1 ohms maximum

3.7 Primer deterioration. - See para. 6.6.

3.8 Benite. The moisture content of the benite at the loading station at the time of loading shall not exceed 1.00 percent.

3.9 Lacquer in flash holes. The lacquer in primer flash holes shall comply with illustration 1, 2, or 3 of Figure 1.

3.10 Workmanship. - The cartridge and all components shall be free from cracks, burrs, chips, dirt, grease, rust, or other foreign material. The cleaning method shall not be injurious to any of the parts; nor shall the parts be contaminated by the cleaning agents. All components shall be free of cracks, seams, splits, cold shuts, inclusions and any other similar defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

a) Quality conformance inspection

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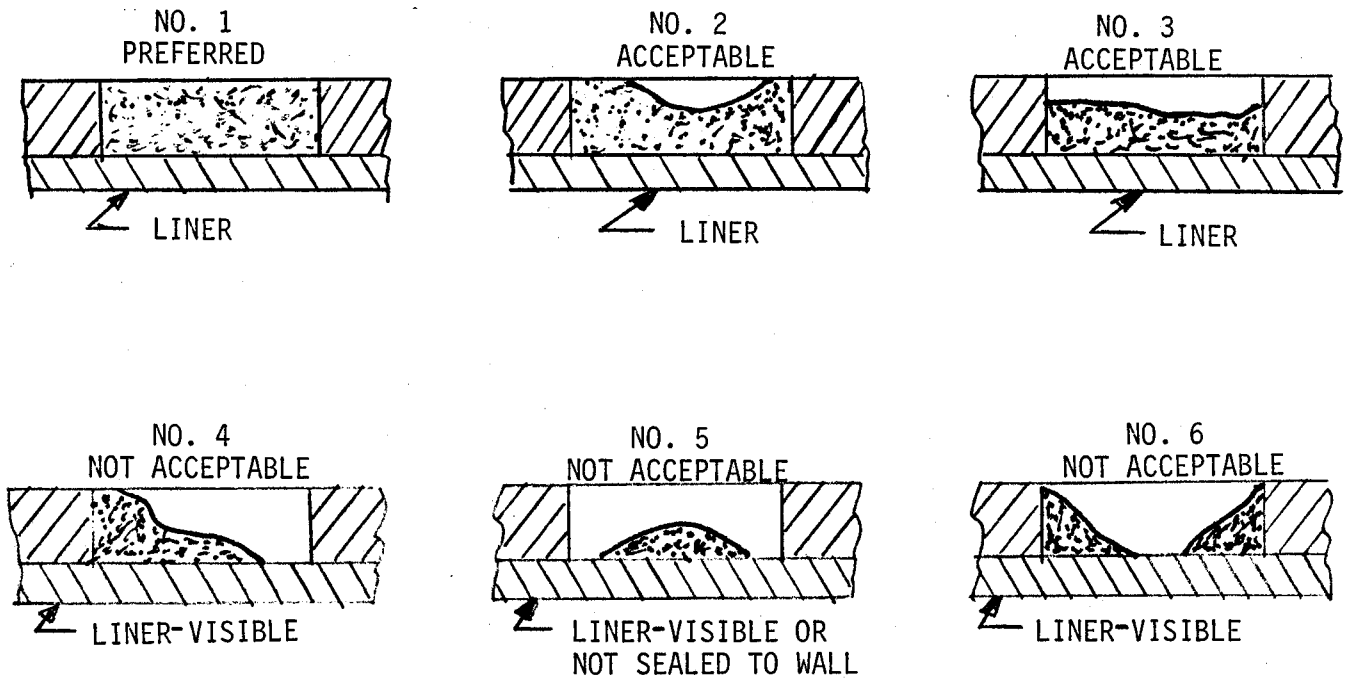


Figure 1. Primers - standards for lacquer conditions in flash holes.

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4.3 Quality conformance inspection.

4.3.1 Lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain a maximum of:

- One lot of propellant
- Combustible cases from one lot interfix from one manufacturer
- Primers from one lot interfix from one manufacturer
- Case base and seal assemblies from one lot interfix number from one manufacturer

4.3.1.1 Lot interfix numbers. - A different cartridge lot interfix number shall be assigned for each slug manufacturer.

4.3.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the classification of defects, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be used if approved by the Government.

b. Interface characteristics. - Interface characteristics identified by the interface symbol on the drawing are identified by an "I" preceding the defect number in the classification of defects and tests.

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QUALITY CONFORMANCE INSPECTION

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Nylon Band				12526296
					NEXT HIGHER ASSEMBLY 12526294
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material	1		3.2	Certification
Minor					
201	Inside Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

DOD-C-70514 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.2	Band Retainer				12526297
					NEXT HIGHER ASSEMBLY 12526294
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threads, maximum		0.40%	3.2	Gage
Minor					
201	Protective finish missing, inadequate or improper		0.65%	3.2	4.4.3.2.1
202	Pitch diameter of threads, minimum		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Outside diameter		0.65%	3.2	Gage
205	True position of slots to minor diameter of thread		0.65%	3.2	Gage
206	True position of O.D. to minor thread diameter		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.3	Body - After Heat Treat Configuration				12526295
					NEXT HIGHER ASSEMBLY 12526294
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Tensile strength	3		3.2	4.4.3.1.2
102	Yield strength	3		3.2	4.4.3.1.2
103	Elongation	3		3.2	4.4.3.1.2
	Information				
	Hardness		100%	3.2	4.4.3.1.1

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.2.4	Body				12526295
					NEXT HIGHER ASSEMBLY 12526294
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of crack or surface defects		100%	3.2	4.4.3.1.3
102	Pitch diameter of adapter threads, minimum		0.40%	3.2	Gage
103	Diameter of front bourrelet		0.40%	3.2	Gage
Minor					
201	Runout of adapter threads		0.65%	3.2	Gage
202	Runout of band retainer threads		0.65%	3.2	Gage
203	Pitch diameter of adapter threads, maximum		0.65%	3.2	Gage
204	Pitch diameter of band retainer threads, minimum		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12526678
4.3.2.4	Spring Disc				NEXT HIGHER ASSEMBLY 12526679
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Stock thickness		0.65%	3.2	Gage
203	Height of formed part		0.65%	3.2	Gage
204	Hardness		1/	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
1/ Three pieces per heat treat lot. No defects allowed.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12526294
4.3.2.5	Super Slug Body Assembly				NEXT HIGHER ASSEMBLY 12526309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Unit weight		100%	3.2	Scale
102	Large diameter on band		0.40%	3.2	Gage
103	Small diameter on band		0.40%	3.2	Gage
104	Band integrity		100%	3.2	4.4.3.1.4
105	Band seating secure	2/Lot		3.2	Visual
Minor					
201	Runout of large diameter on band		0.65%	3.2	Gage
202	Band groove angle		0.65%	3.2	Gage
203	Band groove location		0.65%	3.2	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.3.2.5	Super Slug Body Assembly				12526294
					NEXT HIGHER ASSEMBLY 12526309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
204	Surface protection missing or defective		0.65%	3.2	4.4.3.2.2
205	Band/Band seat cleanliness prior to assembly		0.65%	3.2	Visual
206	Lot markings missing, incorrect or illegible		0.65%	3.2	Visual
207	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.2.6	Adapter				12526298
					NEXT HIGHER ASSEMBLY 12527161
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of ring threads, minimum		0.40%	3.2	Gage
102	Pitch diameter of body threads, maximum		0.40%	3.2	Gage
Minor					
201	Protective finish missing, inadequate or improper		0.65%	3.2	4.4.3.2.1
202	Pitch diameter of ring threads, maximum		0.65%	3.2	Gage
203	Pitch diameter of body threads, minimum		0.65%	3.2	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.3.2.6	Adapter				12526298
					NEXT HIGHER ASSEMBLY 12527161
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
204	Presence of radius on shoulder		0.65%	3.2	Visual
205	Runout of body threads		0.65%	3.2	Gage
206	Length to front shoulder		0.65%	3.2	Gage
207	Runout of ring threads to front face		0.65%	3.2	Gage
208	Runout of ring threads to shoulder surface		0.65%	3.2	Gage
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12527161
4.3.2.7	Case Cap Assembly				NEXT HIGHER ASSEMBLY 12526250
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ring torque not correct	1/		3.2	4.4.2.3
102	Component missing or improperly assembled		100%	3.2	Visual
103	Ring above flush with front face of adapter		100%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

1/ One sample per half shift, no defects allowed

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.8	Spring Disc				12526678
					NEXT HIGHER ASSEMBLY 12526679
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Stock thickness		0.65%	3.2	Gage
203	Height of formed part		0.65%	3.2	Gage
204	Hardness		1/	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
1/ Three pieces per heat treat lot. No defects allowed.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.9	Ring				12526280
					NEXT HIGHER ASSEMBLY 12527161
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of adapter threads, maximum		0.40%	3.2	Gage
Minor					
201	Protective finish missing, inadequate, or improper		0.65%	3.2	4.4.3.2.1
202	Pitch diameter of adapter threads, minimum		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Width of wrench slots		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12526679
4.3.2.10	Propelling Charge Case Assembly				NEXT HIGHER ASSEMBLY 12526250
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parts missing or improperly assembled		100%	3.2	Visual
102	Retaining ring not seated in base case groove		100%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.11	Primer, Electric, Assembly for 120mm Super Slug	1 OF 1	12527160
			NEXT HIGHER ASSEMBLY 12526250

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charge weight not correct		100%	3.3	Scale
102	Disassembly torque not correct		0.40%	3.2	4.4.2.3
103	Liner improperly positioned (prior to loading)		100%	3.2	Visual
104	Flash hole sealing improper (prior to loading)		100%	3.9	Visual
105	Lacquer missing from plug threads		100%	3.2	Visual
106	Sealant missing from body threads		100%	3.2	Visual
107	Moisture content of Benite	1/		3.8	4.4.1.2
Minor					
201	Overall length, max		0.65%	3.2	Scale
202	Marking improper		0.65%	3.2	Visual
203	Depth to plug		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

1/ One 100 gram sample of benite shall be selected at the loading station, during each 8 hours production. If the moisture content of the sample exceeds the applicable requirement the primers loaded since the last acceptance sample shall be rejected.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.3.2.12	Propulsion/Ignition Assembly (LAP Inspection)				12526250
					NEXT HIGHER ASSEMBLY 12526309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Adhesive joint incomplete or not smoothly contoured		100%	3.2	Visual
102	Propellant charge weight not correct		100%	3.2	4.4.1
103	Minimum strength requirements not met on adhesive joint	4.4.2.1		3.2	4.4.2.1
104	Cartridge case base coat thickness not correct	4.4.2.2		3.2	4.4.2.2
105	Cartridge case finish coat thickness not correct	4.4.2.2		3.2	4.4.2.2
106	Missing or damaged primer "O" Ring		100%	3.2	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.3.2.12	Propulsion/Ignition Assembly (LAP Inspection)				12526250
					NEXT HIGHER ASSEMBLY 12526309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107	Primer torque not correct (prior to staking)	1/		3.2	4.4.2.3
108	Primer depth below surface of base case not correct		100%	3.2	Gage
109	Primer electrode to primer body resistance not correct (prior to assembly to case base)		100%	3.6	4.4.5
110	Transfer resistance from primer body to case base not correct (prior to staking)		100%	3.6	4.4.5
111	Missing, damaged or improperly positioned parts		0.40%	3.2	Visual

NOTES:
1/ Two samples per shift with zero defects

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 12526250
4.3.2.12	Propulsion/Ignition Assembly (LAP Inspection)				NEXT HIGHER ASSEMBLY 12526309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Case coating peeling		100%	3.2	Visual
113	Non-uniform adhesive application on case cap		100%	3.2	Visual
114	Non-uniform adhesive application on cartridge case		100%	3.2	Visual
I 115	Cartridge does not chamber		100%	3.2	4.4.4.1
116	Primer stake missing		100%	3.2	Visual
Minor					
201	Cartridge marking incorrect, missing or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12526248
4.3.2.13	Seal				NEXT HIGHER ASSEMBLY 12526654
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material improper	1		3.1	4.4.3.3
Minor					
201	Die/mold check, each cavity		0.65%	3.2	4.4.3.4.1
202	Shore hardness after molding		0.65%	3.2	4.4.3.3.3
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.14	Plug, Threaded	1 OF 1	12526689
			NEXT HIGHER ASSEMBLY 12526250

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threads, min		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.15	Cartridge, 120mm Super Slug (Proving Ground LAP)				12526309
					NEXT HIGHER ASSEMBLY: Final
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cartridge does not chamber		100%	3.2	4.4.4.2
102	Overall length, max		100%	3.2	Gage
103	PI system/body assembled hand tight		100%	3.2	Manual
104	Absence of adhesive		100%	3.2	Visual
105	Missing Seal		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.16	Cartridge, 120mm Super Slug (Performance Test)				12526309
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		10			
101	Chamber Pressure			3.4.1	4.4.6.3.1
102	Muzzle Velocity			3.4.2	4.4.6.3.2 Info Only
103	Recoil Momentum			3.4.3	4.4.6.3.3
104	Misfire			3.4.4	4.4.6.3.4
105	Metal Parts Security			3.4.5	4.4.6.3.5
106	Case Base Extraction			3.4.6	4.4.6.3.6
Minor	None defined				

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.17	Packing and Marking for Propulsion/Ignition Assembly, 120mm Super Slug (packaged and sealed item in fiber container in wood box outer pack)				12527256
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fiber container cut or damaged completely through		0.40%	3.2	Visual
102	Loose or damaged end of fiber container		0.40%	3.2	Visual
103	Loose contents		0.40%	3.2	Visual
104	Missing or improperly applied tape on fiber container joint		0.40%	3.2	Visual
Minor					
201	Damage to outer layers of fiber container		0.65%	3.2	Visual
202	Marking missing, illegible or incorrect		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

DOD-C-70514 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.18	Unitization Procedures for Boxed 120mm Tank Ammunition and Components on 4-Way Entry Pallets				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Missing or loose strapping		0.40%	3.2	Visual - Manual
102	Strapping improperly positioned		0.40%	3.2	Visual
103	Incorrect number of packed boxes on pallet		0.40%	3.2	Visual
104	Boxes not all positioned in correct orientation		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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4.3.3 Testing. - Testing is described in the Quality Conformance Inspection tables.

4.3.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4 Methods of inspection.

4.4.1 Propellant.

4.4.1.1 Propellant weight. - The propellant shall be weighed and check weighed 100%. Check weighing shall be a completely independent operation from the original weighing using a different scale and, if performed manually, by a different operator. Any propellant charge that does not meet applicable weight requirement shall be classified defective and removed from the lot.

4.4.1.2 Benite. - The moisture content of the Benite shall be determined in accordance with MIL-STD-286, Method 102.1.

4.4.2 Cartridge case.

4.4.2.1 Adhesive joint strength test. - The integrity of the adhesive joint shall be verified. The selected case cap and cartridge case shall be attached to the appropriate test fixtures (see Figures 2 and 3) and introduced into the production line for adhesive application and assembly in accordance with standard production methods. Frequency of test shall be at a minimum of one (1) every twenty-five (25) propulsion/ignition assemblies. After three (3) consecutive acceptable tests at this frequency, the frequency may be decreased to one (1) every fifty (50) propulsion/ignition assemblies. After three (3) consecutive acceptable tests at this frequency, the frequency may be decreased to one (1) every one-hundred (100) propulsion/ignition assemblies. In the event of a failure, frequency shall revert back to one (1) every twenty-five (25) propulsion/ignition assemblies and all product produced since the last acceptable verification shall be rejected. Whenever the gluing of 120mm combustible cartridge cases has been interrupted for a period of ninety (90) working days, the test frequency shall revert back to one (1) sample every twenty-five (25) cartridges.

4.4.2.1.1 Adhesive cure. - The test cartridge case shall be cured for two (2) hours minimum at ambient temperature prior to test.

4.4.2.1.2 Test of bond. - The test cartridge shall be attached to a tensile test fixture and a 8900 N minimum load shall be applied at a maximum rate of 45 + 5 mm/minute with a dwell of 5 seconds minimum at full load. Any loosening of the bond shall be considered a failure.

4.4.2.2 Cartridge case coating. - The cartridge case coating requirement shall be verified by introducing a test vehicle into the cartridge coating

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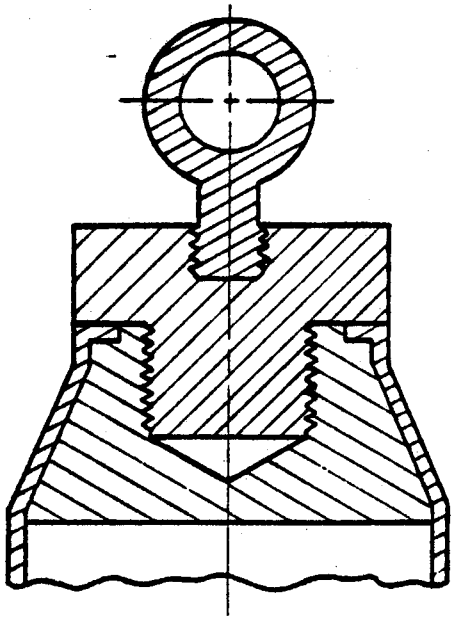


Figure 2. Case cap attachment fixture.

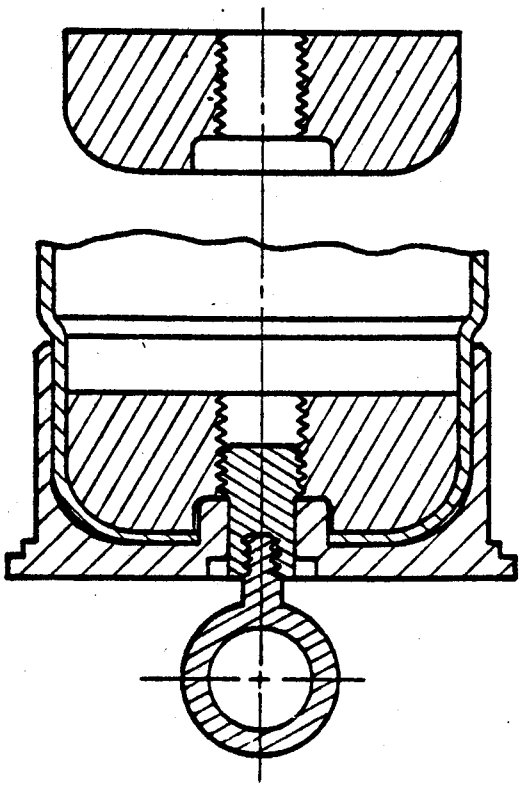


Figure 3. Case base attachment fixture.

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operation and subjecting it to the same coating process as applied to cartridges in production. Frequency of test shall be a minimum of one (1) every twenty-five (25) cartridges or one (1) every hour whichever results in the greater number of samples. After three (3) consecutive acceptable checks have been made, frequency may decrease to one (1) every four hours. In the event of a failure, frequency shall revert back to one (1) every twenty-five (25) cartridges or one every hour and all product coated since the last acceptable verification shall be rejected. The first assembly shall be checked at the start of each shift and after a rejection.

4.4.2.2.1 Peeling. - The cartridge case shall be visually examined for evidence of peeling or delamination using samples prepared per paragraph 4.4.2.2. Evidence of peeling shall be cause for rejection.

4.4.2.3 Torque test. - The minimum assembly torque, as specified in the applicable drawing, shall be gradually applied to the applicable joint in the direction required for assembly. Any perceptible movement between the interfacing units shall be cause for rejection of all assemblies produced since the last acceptable check.

4.4.3 Projectile.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness. - Brinell or Rockwell Hardness Test shall be performed on heat treatment lots in the locations defined by Figure 4. ASTM Method E18 shall be used for performing Rockwell Hardness. Readings shall be taken in the designated area at two (2) locations approximately 180 degrees apart. Prior to performing the test, the area in which the readings are to be taken shall be cleaned of scale or other foreign matter. Two (2) samples with the lowest hardness readings and the one (1) sample with the highest reading shall be used for performing the tension tests. ASTM Method E10 for Brinell Hardness may be used as an alternate test method.

4.4.3.1.2 Tension. - From the samples provided by 4.5.3.1.1 tension tests shall be performed utilizing the equipment and procedures prescribed by ASTM E8. Yield strength shall be determined by the offset method using a limiting set of 0.1 percent offset. The largest obtainable standard round specimen as prescribed by ASTM E8 shall be taken in the location shown in Figure 3. Failure to comply with the requirements defined in the drawing shall be cause for rejection of the lot. At the option of the contractor, the entire lot may be tempered, reheat treated, and retested; or a new sample may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be tempered, reheat treated and retested. Only one reheat treat cycle is allowed to achieve conformance.

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Location of the test zones for determining tensile strength, yield strength, elongation and hardness

Location of the test zones for determining tensile strength, yield strength, elongation and hardness

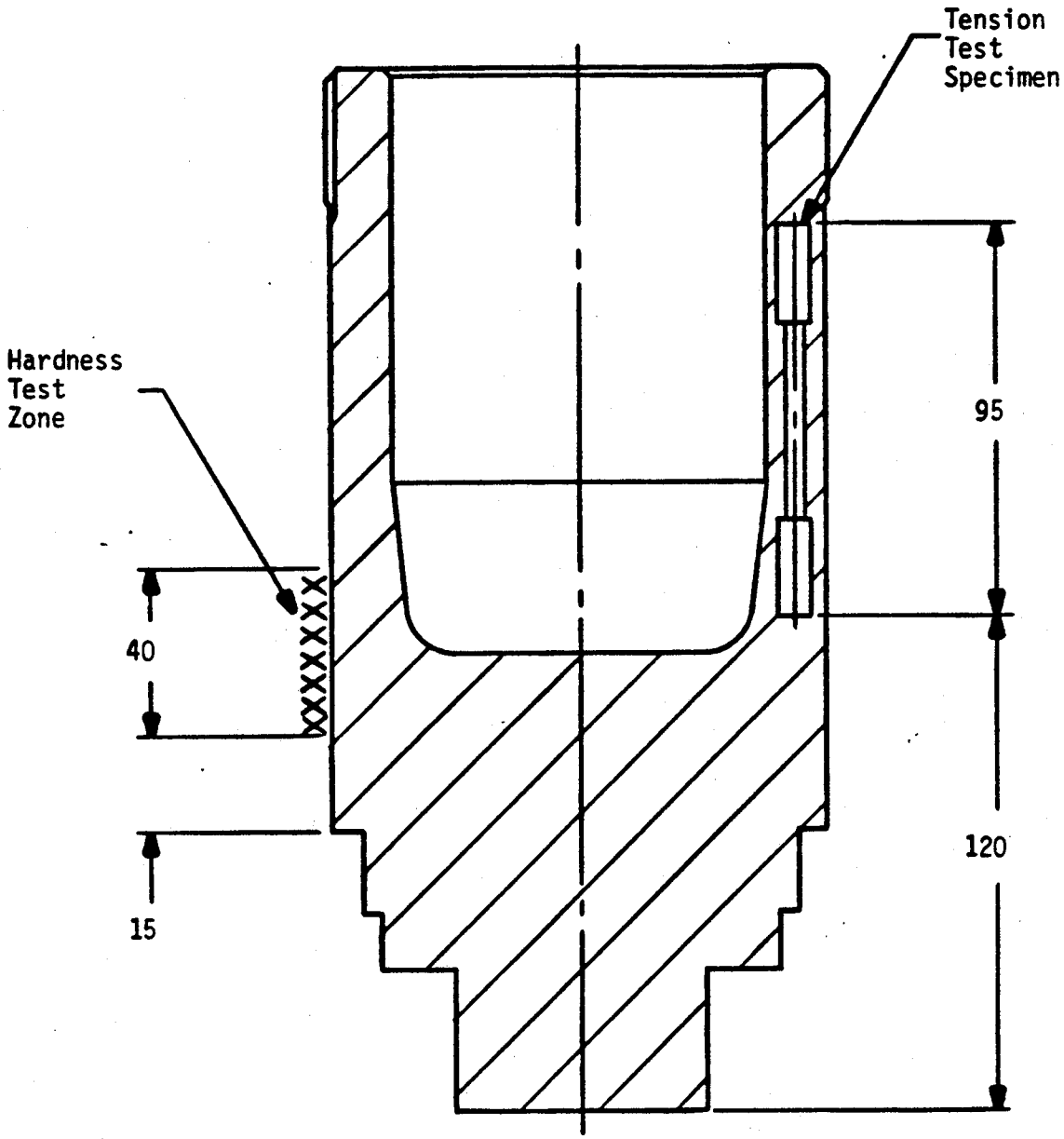


Figure 4. Body.

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4.4.3.1.3 Structural integrity. - Each body, after final machining and prior to application of protective finish, shall be visually examined under 4x magnification on interior and exterior surfaces. Accept criteria shall be as determined by the visual standards of 6.8. Rejected bodies shall be removed from the lot.

4.4.3.1.4 Band integrity. - The band shall be inspected after machining for evidence of cracked or loose bands, using 4X magnification. Rejected items shall be removed from the lot.

4.4.3.2 Protective finish.

4.4.3.2.1 Thickness of protective finish. - The protective finish thickness as defined by the drawing shall be measured using equipment capable of performing the test accurately.

4.4.3.2.2 Paint thickness. - The paint thickness on the body and band assembly shall be measured. Failure to meet the paint thickness requirement shall be cause for rejection of the lot. Frequency of test shall be a minimum of one every twenty-five items. After three consecutive acceptable tests, the frequency may be decreased to one every fifty units. After three consecutive acceptable tests at this frequency, the rate may be decreased to one every one-hundred units. In the event of a failure, the frequency shall revert back to one every twenty-five units and all units since the last acceptable test shall be rejected.

4.4.3.3 Seal examination.

4.4.3.3.1 Specific gravity. - Specific gravity of the rubber blend shall be performed on the sample size specified per ASTM D792. Failure to meet the drawing requirements shall reject the lot.

4.4.3.3.2 Shore hardness of seal rubber blend. - Shore hardness testing shall be performed on the sample size specified per ASTM-D-2240, (Average of 3 readings at least 50mm apart) and shall meet the requirements of Drawing 12524837.

4.4.3.3.3 Shore hardness of seal after molding. - To verify proper curing of the rubber, shore hardness shall be taken on the seal. The hardness shall not be more than 3 points higher than, nor lower than, the value obtained on the test specimen as defined in ASTM D2240 using the same catalyzed batch of rubber used to mold the seal. Failure to meet this requirement shall reject the lot. The reading shall be taken in an area which provides sufficient rubber thickness as not to provide an erroneous hardness reading. The average of three readings at approximately 120° apart shall be used to verify conformance.

4.4.3.4 Dimensional control of molded parts. - In place of the normal sampling associated with the classification of defects, and subject to approval of the Government Contracting Officer, the optional sampling plans in paragraph 4.4.3.6.1 may be used for dimensional control of molded parts. Use of these optional plans is limited to those dimensions and features controlled by the mold cavity. For all specified tests, visual examinations, and

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examinations of dimensions or features not developed exclusively by the mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejection, and subsequent to the last acceptable sample, shall require approval of the Government procuring agency.

4.4.3.4.1 Molded parts. - A sample of three parts (as molded) qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities removed from production because of some fault may, after reworking, be returned to production, providing they pass the qualification test above. A minimum of 4 hours after molding shall elapse before performing inspection.

4.4.4 Chamberability.

4.4.4.1 Propulsion ignition assembly. - The Propulsion/Ignition Assembly, assembled with a Warhead Simulator, shall be inserted into the chamber gage, 12524967. Failure of the assembly to completely enter the gage, when inserted with a maximum force of 27.5 Kg, shall be cause for rejection of the Propulsion/Ignition Assembly.

4.4.4.2 Cartridge. - The cartridge shall be inserted into the chamber gage, 12524967. Failure of the cartridge to completely enter the gage, when inserted with a maximum force of 27.5 kg, shall be cause for rejection of the cartridge.

4.4.5 Primer circuit electrical tests. - The electrical resistance of the cartridge/primer circuit shall be verified. Primer electrode to primer body resistance shall be measured prior to assembling into the propulsion/ignition assembly. Transfer resistance between the primer body and case base shall be measured after the primer has been torqued into the case base but prior to staking it. Resistance shall be measured using instrumentation that will limit test current per drawing requirement. Failure to meet the specified requirement shall be cause for rejection of the assembly.

4.4.6 Performance tests. - Performance tests shall be conducted on test rounds assembled from parts and assemblies which have been inspected in accordance with the applicable Quality Conformance Inspection tests specified in this document.

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4.4.6.1 Test conditions.

4.4.6.1.1 Weapon. - Unless otherwise specified all performance tests shall be conducted using an M256 gun on a fixed mount.

4.4.6.1.2 Temperature. - Unless otherwise specified performance tests shall be conducted with ammunition conditioned at $21^{o}C \pm 1.5^{o}C$ for a period of 24 hours minimum prior to firing.

4.4.6.2 Test procedures.

4.4.6.2.1 Chamber pressure. - Chamber pressure shall be measured piezoelectrically at a location 8.9 cm from the rear face of the tube. The gage shall be positioned at 30° counterclockwise from the vertical center line.

4.4.6.2.2 Retest. - An additional sample the same size as the original sample shall be provided for retest. All retests are at the discretion of the Government Contracting Officer.

4.4.6.3 Performance characteristics

4.4.6.3.1 Chamber pressure. - Chamber pressure at $21^{o}C$ shall be verified piezoelectrically. Failure to meet the specified requirement shall be cause for rejection.

4.4.6.3.2 Muzzle velocity. - Muzzle velocity shall be measured if the Vinti method is used to verify the recoil momentum.

4.4.6.3.3 Recoil momentum. - Recoil momentum shall be verified by a computer routine derived from the Vinti method based on ATI 139981. The calculation approach and the applicable numerical constants and physical descriptions of the gun, propellant and combustible case are given in Appendix A. The actual computer program is attached as Appendix B. Recoil momentum may be optionally verified using a direct recoil measuring technique.

4.4.6.3.4 Misfire. - All firing tests shall be monitored for evidence of misfire. Any misfire shall be cause for rejection.

4.4.6.3.5 Metal parts security. - Metal parts security shall be verified during gun firing tests. Verification will be accomplished by placement of a smear camera approximately 20 meters from muzzle. Any projectile that exhibits evidence of breakup shall be cause for rejection.

4.4.6.3.6 Extraction. - The case base and seal assembly shall be extracted from the chamber after firing. (See 6.5.) Any case base and seal assembly that fails to extract shall be cause for rejection.

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5. PACKAGING

5.1 Propulsion ignition assembly.

5.1.1 Cleaning. - Item shall be cleaned in accordance with process C-1 of MIL-P-116, if required.

5.1.2 Drying. - Item shall be dried in accordance with MIL-P-116, if required.

5.1.3 Preservation. - Not applicable.

5.1.4 Unit and intermediate packaging

5.1.4.1 Level A. - To be determined.

5.1.4.2 Level B. - To be determined.

5.1.4.3 Level C. - Two each, 120mm, Super Slug Propulsion Ignition Assemblies shall be packaged in a spiral wound fiber container, 12527252, in accordance with 12527256.

5.1.5 Packing.

5.1.5.1 Level A. - To be determined.

5.1.5.2 Level B. - To be determined.

5.1.5.3 Level C. - Two each package from para. 5.1.4 shall be packed into a wood box, 12527251, in accordance with 12527252.

5.1.6 Marking.

5.1.6.1 Level A. - To be determined.

5.1.6.2 Level B. - To be determined.

5.1.6.3 Marking to be in accordance with dwg. 12527256.

5.1.7 Unitization.

5.1.7.1 Level A. - To be determined.

5.1.7.2 Level B. - To be determined.

5.1.7.3 Level C. - The packs in para 5.1.5.3 may be utilized on a pallet base and secured with steel strapping so that safety and structural integrity of each pack as well as the entire unit load is not compromised.

5.2 Projectile.

5.2.1 Cleaning. - Item shall be cleaned in accordance with process C-1 of MIL-P-116, if required.

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5.2.2 Drying. - Item shall be dried in accordance with MIL-P-116, if required.

5.2.3 Preservation. - Not applicable.

5.2.4 Unit and intermediate packaging.

5.2.4.1 Level A. - The items listed in para. 5.2 shall be unit and intermediate packaged in accordance with Level A of MIL-STD-1169. In addition, each item shall be separated by appropriate partitioning and supported by appropriate cushioning to prevent physical damage. Also, each unit or intermediate package shall be enclosed in a barrier bag in accordance with Type I, Class E of MIL-B-117 to prevent moisture damage (corrosion). Evacuate air in barrier bag prior to heat sealing.

5.2.4.2 Level B. - Items listed in para. 5.2 shall be unit and intermediate packaged in accordance with Level B of MIL-STD-1169. In addition, each item shall be separated by appropriate partitioning and supported by appropriate cushioning to prevent physical damage. Also, each unit or intermediate package shall be enclosed in a barrier bag in accordance with Type I, Class E of MIL-B-117 to prevent moisture damage (corrosion). Evacuate air in bag prior to heat sealing.

5.2.4.3 Level C. - The items listed in para. 5.2 shall be unit and intermediate packaged in accordance with Level C of MIL-STD-1169. In addition, each item in package shall be separated by appropriate partitioning and supported by appropriate cushioning to prevent physical damage. Also, each unit or intermediate package shall be enclosed in a barrier bag in accordance with Type I, Class C of MIL-B-117 to prevent moisture damage (corrosion). Evacuate air in barrier bag prior to sealing.

5.2.5 Packing.

5.2.5.1 Level A. - The package in para. 5.2.4.1 shall be packed in accordance with Level A of MIL-STD-1169. Sufficient filler shall be added to pack to prevent tearing of bag and insure a tight pack.

5.2.5.2 Level B. - The package in para. 5.2.4.2 shall be packed in accordance with Level B of MIL-STD-1169. Sufficient filler shall be added to pack to prevent tearing of bag and insure a tight pack.

5.2.5.3 Level C. - The package in para. 5.2.4.3 shall be packed in accordance with Level C of MIL-STD-1169. Sufficient filler shall be added to pack to prevent tearing of bag and insure a tight pack.

5.2.5.4 Reuse of packing materials. - Reuse of all packing material is permissible and recommended when considered economically feasible by the procuring activity and the material is in reusable condition.

5.2.5.5 Advanced packing technology. - Contractors are encouraged to use advanced packing technology and innovative methods or materials for the purpose of effecting packing economies provided materials meet the required specifications.

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5.2.6 Marking.

5.2.6.1 Level A. - Marking shall be in accordance with MIL-STD-1169.

5.2.6.2 Level B. - Marking shall be in accordance with MIL-STD-1169.

5.2.6.3 Level C. - Unless otherwise specified, unit, intermediate (when provided), and exterior packs shall, as a minimum, be marked as follows by any means which provides legibility and durability:

A. Noun Nomenclature
B. Quantity per Carton
C. Contract No.
D. Seller's Name & Address
E. Lot No. (13 digit code per MIL-STD-1168)
F. Date of Mfg. (Mo. & Yr.)
G. Carton ___ of ___ Cartons
H. Carton/Package Wt.
I. Army Drawing & Revision Part No.

5.3 Shipping.

5.3.1 Super slug body assembly. - When super slug body assemblies from more than one lot are shipped at the same time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of lots in transit.

5.3.2 Propulsion, ignition assembly. - When propulsion ignition assemblies loaded with propellant from more than one lot are shipped at one time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of lots in transit.

5.3.3 Seal. - Seals may be shipped separately from the body assembly and the propulsion ignition assembly. They shall be packaged in accordance with level C of MIL-STD-1169.

5.3.4 Propellant. - When propellant from more than one lot is shipped at one time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. - The item covered by this specification is intended for use in the M256 gun for proofing recoil systems and gun/mount combinations for the M1A1 Tank.

6.2 Ordering data. - See MIL-A-48078

6.2.1 Data requirements. - When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract.

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Paragraph	Data Requirement	Applicable DID
6.8	Ammunition Data Cards	DI-L-1410

Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

6.3 Submission of inspection equipment for design approvals. - See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Munitions and Chemical Command, Attn: AMSMC-QAT-I(D), Dover, NJ 07801.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or US Army Armament Research and Development Command drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Extraction. - The term extraction shall mean that the fired case base and seal assembly is cleared from the rear of the weapon by the automatic mechanism.

6.6 Deterioration of primer. - If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into propulsion ignition assemblies exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to, and must satisfactorily pass, the check test for deterioration in accordance with MIL-P-70451 immediately before the primer lot is assembled into the propulsion ignition assemblies. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the primer check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the propulsion ignition assemblies, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Definition of packaging and packing levels

6.7.1 Level A. - Full military protection, for unrestricted (OCONUS) shipment and storage.

6.7.2 Level B. - Limited military protection, for limited storage and (OCONUS) shipment.

6.7.3 Level C. - For interplant (CONUS) shipment and very limited storage.

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6.7.4 For further clarification of packaging levels see AR-700-15.

6.8 Submission of ammunition data cards. - In addition to the normal distribution of records when this cartridge is procured by the U.S. Army Armament, Materiel and Chemical Command, one copy of all ammunition data cards shall be forwarded to each of the following: Commander, ARDC, ATTN: SMCAR-LCU-M and AMSMC-QAT-A(D), Dover, NJ 07801-5001; and, Commander, AMCCOM, ATTN: AMSMC-QAD(R), Rock Island, IL 61299-6000.

Custodian
Army-AR

Preparing Activity
Army-AR

Project Number: 1315-A641

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APPENDIX A

RECOIL MOMENTUM CALCULATION METHOD

10.0 SCOPE

10.1 Scope.

This method of computing recoil momentum is based upon BRL Report #538. (The Effective Velocity of Escape of the Powder Gas from a Gun, John P. Vinti, dated 2 April 1945.) The propellant gas for the 120mm is modified by the combustible case. Input to the standard Vinti equations are modified as outlined below to account for effects of the combustible case.

Specific force and flame temperature of the effective propellant are computed as a weighted average of the M30 propellant and the cartridge case. Properties to be used are listed below:

Propellant	Flame Temp (^{o}K)	Spec. Force (ft lb/lb.)	Gamma	Weight
M30	3040	364,000	1.24	Varies
Case	1611	192,000	1.258	.825 (German Average
			.79	(U.S. Averag

Inputs to the Vinti equations.

Title	Units	Symbol	120mm Value
Muzzle Velocity	Ft./sec	V	Note 1
Bore Area	Square Inches	A	17.53127
Chamber Volume	Cubic Inches	C	612
Weight Projectile	Pounds	W1	Note 2
Weight Propellant	Pounds	W2	Note 3
Web Size	Inches	W	0.060 (For M-30)
Specific Force	Ft. lb/lb.	L	Note 4
Gamma	--	G	Note 5
Flame Temp.	Degrees Kelvin	TØ	Note 6

Note 1: This is a measured value. Units used are ft/sec for the Vinti formulas. The HP41 program listed in Appendix B expects velocity in ft/sec for the general case. For 120mm application, Flag 1 is set, and input is in meters/second. (Converted by program for computation.)

Note 2: Input is in pounds for general case and in Kg for 120mm (Flag 1 set). Internal conversion is made by the program for computation.

Note 3: Input in pounds for general case. For 120mm input is in Kg. With Flag 1 set, weight of case is added and converted to pounds by the program for computation.

Note 4: Input is in ft lbs/lb. in the general case. For 120mm, a weighted average is computed by the program.

Note 5: Input is dimensionless. For 120mm a weighted average is computed.

Note 6: Input is in degrees Kelvin in the general case. For 120mm a weighted average is computed.

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APPENDIX A

RECOIL MOMENTUM CALCULATION METHOD

With the input as listed above, the following sequences of assignment statements will compute escape velocity (in ft/sec) and impluse (in lb-sec.) The HP41 program with Flag 1 set lists impulse in Newton seconds, rounded to the closest five.

The following table is a list of symbols and assignment statements that define an algorithm for the Vinti method. Computation lines and memory assignments for the HP41 program are included.

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RECOIL MOMENTUM CALCULATION METHOD

COMP #'s	MEM #'s	SYMBOL
	01	V
	2	A
	3	C
	4	W1
	5	W2
	6	W
	7	L
	8	G
	9	TØ
51 - 57	10	D1 = W2/C/.036127
58 - 64	11	D = 2*SQR(A/#PI)
65 - 80	12	K = 25.8*W/D/D1/SQR(.001*TØ-1)
81 - 85	13	N = 1/(G-1)
86 - 89	14	E = W2/W1
90 - 97	15	E1 = E/2(N + 1)
98 - 118	16	B1 = .75*(SQR(1+8*N*E1/3)-1)/N
119 - 135	17	E2 = E1-4*N*(N-1)*(B1 3)/15
136 - 156	18	B = .75*(SQR(1+8*N*E2/3)-1/N
157 - 162	19	AØ = B/(1+B)
163 - 177	20	D2 = 1/((2*N+3)/(1/AØ-1/E1))
178 - 197	21	H = 1-N*AØ/6+N*(N-6)*(AØ 2)/90
198 - 217	22	F = (((G+1/2) ((3-G)/2(G-1)))/SQR(G)
218 - 241	23	R1 = L*32.17-(G-1)*(1+K+E*D2)*(V 2)/2/E
242 - 269	24	V1 = (H/2+H*SQR((1+E/2)/(1+E/3))*F*SQR(R1/V)*V
270 - 279	25	R = (W1*V+W2*V1)/32.17
	26	Case wt. (lb)
	27	Change wt. (lb)
	28	Specific force change (ft lb/lb)
	29	Flame temp. change $^{\circ}$K
	30	Gamma change

V1 is escape velocity in ft/sec. (Output is R-24)
R is impluse (recoil) in lb-sec. (Output is R-25)
Output for 120 program (Flag 1 set) is
IMP = _____ (in Newton-seconds) (Impulse)

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APPENDIX B

RECOIL MOMENTUM COMPUTER PROGRAM

10.0 SCOPE

10.1 Scope.

```
PRP ""
01♦LBL "VINTI"
02 "120 ?-SF1"
03 PROMPT
04 FS? 01
05 GTO 02
06 "VM= "
07 PROMPT
08 STO 01
09 ARCL X
10 PRA
11 "BA= "
12 PROMPT
13 STO 02
14 ARCL X
15 PRA
16 "CHV= "
17 PROMPT
18 STO 03
19 ARCL X
20 PRA
21 "WP= "
22 PROMPT
23 STO 04
24 ARCL X
25 PRA
26 "WC= "
27 PROMPT
28 STO 05
29 ARCL X
30 PRA
31 "WEB= "
32 PROMPT
33 STO 06
34 ARCL X
35 PRA
36 "FORCE= "
37 PROMPT
38 STO 07
39 ARCL X
40 PRA
41 "GAMMA= "
42 PROMPT
43 STO 08
44 ARCL X
45 PRA
46 "FL T= "
47 PROMPT
48 STO 09
49 ARCL X
50 PRA
51♦LBL 01
52 RCL 05
53 RCL 03
54 /
55 .036127
56 /
57 STO 10
58 RCL 02
59 PI
60 /
61 SQRT
62 2
63 *
64 STO 11
65 RCL 09
66 .001
67 *
68 1
69 -
70 SQRT
71 RCL 10
72 *
73 1/X
74 25.8
75 *
76 RCL 06
77 *
78 RCL 11
79 /
80 STO 12
81 RCL 08
82 1
83 -
84 1/X
85 STO 13
86 RCL 05
87 RCL 04
88 /
89 STO 14
90 RCL 14
91 2
92 /
93 RCL 13
94 1
95 +
96 /
97 STO 15
98 3
99 ENTER↑
100 4
101 /
102 RCL 13
103 /
104 8
105 ENTER↑
106 3
107 /
108 RCL 13
109 *
110 RCL 15
111 *
112 1
113 +
114 SQRT
115 1
116 -
117 *
118 STO 16
119 RCL 15
120 4
121 ENTER↑
122 15
123 /
124 RCL 13
125 *
126 RCL 13
127 1
128 -
129 *
130 RCL 16
131 3
132 Y↑X
133 *
134 -
135 STO 17
136 3
137 ENTER↑
138 4
139 /
140 RCL 13
141 /
142 8
143 ENTER↑
144 3
145 /
146 RCL 13
147 *
148 RCL 17
149 *
150 1
```


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APPENDIX B

RECOIL MOMENTUM COMPUTER PROGRAM

```
151 +
152 SQRT
153 1
154 -
155 *
156 STO 18
157 RCL 18
158 RCL 18
159 1
160 +
161 /
162 STO 19
163 1
164 ENTER↑
165 RCL 13
166 2
167 *
168 3
169 +
170 /
171 RCL 19
172 1/X
173 RCL 15
174 1/X
175 -
176 *
177 STO 20
178 RCL 19
179 RCL 13
180 *
181 6
182 /
183 CHS
184 1
185 +
186 RCL 13
187 90
188 /
189 RCL 13
190 6
191 -
192 *
193 RCL 19
194 X↑2
195 *
196 +
197 STO 21
198 RCL 08
199 1
200 +
201 2
202 /
203 3
204 RCL 08
205 -
206 RCL 08
207 1
208 -
209 2
210 *
211 /
212 Y↑X
213 RCL 08
214 SQRT
215 1/X
216 *
217 STO 22
218 RCL 14
219 RCL 20
220 *
221 RCL 12
222 +
223 1
224 +
225 RCL 14
226 /
227 RCL 01
228 X↑2
229 *
230 RCL 08
231 1
232 -
233 2
234 /
235 *
236 CHS
237 RCL 07
238 32.17
239 *
240 +
241 STO 23
242 RCL 14
243 2
244 /
245 1
246 +
247 RCL 14
248 3
249 /
250 1
251 +
252 /
253 SQRT
254 RCL 21
255 *
256 RCL 23
257 SQRT
258 RCL 22
259 *
260 RCL 01
261 /
262 *
263 RCL 21
264 2
265 /
266 +
267 RCL 01
268 *
269 STO 24
270 RCL 04
271 RCL 01
272 *
273 RCL 05
274 RCL 24
275 *
276 +
277 32.17
278 /
279 STO 25
280 RTN
281 0.025
282 ENTER↑
283 FIX 9
284 PRREGX
285 STOP

286*LBL 02
287 FIX 0
288 "VM= "
289 PROMPT
290 ARCL X
291 PRA
292 RCL X
293 .3048
294 /
295 STO 01
296 17.530127
297 STO 02
298 612
299 STO 03
300 FIX 1
```


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APPENDIX B

RECOIL MOMENTUM COMPUTER PROGRAM

```
001 "WP= "
002 PROMPT
003 ARCL X
004 PRA
005 RCL X
006 .45359237
007 /
008 STO 04
009 FIX 3
010 "CASE WT= "
011 PROMPT
012 ARCL X
013 PRA
014 .45359237
015 /
016 STO 26
017 "WCGE= "
018 FIX 2
019 PROMPT
020 ARCL X
021 PRA
022 RCL X
023 .45359237
024 /
025 STO 27
026 RCL 26
027 +
028 STO 05
029 "SP FORCE= "
030 PROMPT
031 STO 28
032 ARCL X
033 PRA
034 "FLAME T= "
035 PROMPT
036 STO 29
037 ARCL X
038 PRA
039 "GAMMA= "
040 PROMPT
041 STO 30
042 ARCL X
043 PRA
044 .06
045 STO 06
046 RCL 05
047 RCL 26
048 -
049 RCL 28
050 *
051 RCL 26
052 192000
053 *
054 +
055 RCL 05
056 /
057 STO 07
058 RCL 30
059 RCL 27
060 *
061 RCL 26
062 1.258
063 *
064 +
065 RCL 05
066 /
067 STO 08
068 RCL 05
069 RCL 26
070 -
071 RCL 29
072 *
073 RCL 26
074 1610
075 *
076 +
077 RCL 05
078 /
079 STO 09
080 XEQ 01
081 RCL 25
082 4.44822
083 *
084 10
085 /
086 FIX 0
087 RND
088 10
089 *
090 "IMP= "
091 ARCL X
092 PRA
093 ADV
094 ADV
095 ADV
096 ADV
097 ADV
098 .END.
```


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